

UI/UX Test-Coverage Audit — Lava :app (client) + :api-app

Date: 2026-06-25 **Scope:** Read-only source inspection of every client feature/* module + the api-app UI, mapped against existing Compose UI Challenge Tests (app/src/androidTest/..., api-app/src/androidTest/...) and per-feature ViewModel/unit tests (feature/*/src/test/, api-app/src/test/). **Posture:** Anti-bluff (§6.AB / §6.J). GREEN = a test that drives the production code path and asserts on user-visible state (rendered text, persisted row, behavior). WEAK = a test exists but asserts presence/classpath only, not behavior. GAP = no UI test for that surface.

Honesty note: "GREEN" here means the test is *written* to assert user-visible state. Per the repo-wide §6.X-debt, the :app client Challenge Tests have NOT been executed on an emulator on the current darwin/arm64 host (only the api-app tests use test-tags and have a documented run path). This audit rates the *design quality* of the assertions from source, not a green run. Where execution status matters it is called out.

1. Headline coverage stats

Metric	Value
Client feature modules	19
User-facing screens (client)	~22 (incl. onboarding 5 steps as one wizard)
api-app screens	1 (ApiControlScreen)
Distinct high-risk interactive components (toggles/switches/dialogs/text-fields)	~30
Client Challenge Tests (Challenge*Test.kt)	48 files
api-app Challenge Tests	5 files (C01–C05)

Metric	Value
Per-feature ViewModel/unit test files (client)	38
api-app unit test files	9
Features with ≥ 1 GREEN behavior-driving UI test	~9 / 19
Features WEAK only (Class.forName / reachable-marker)	5 (visited, favorites, bookmarks, category, connection)
Features with GAP (no direct UI test)	account, main, search_input, credentials_manager, rating(partial)
% of screens with a behavior-asserting UI test	~45%
% with only WEAK / classpath UI test	~25%
% GAP (no UI test)	~30%

Bottom line: coverage is thin and uneven. Onboarding, search, topic, and the api-app control screen are genuinely well-covered. The high-risk **provider_config toggles/switches/serialization** are covered only at the **unit (ViewModel) layer** — there is **no end-to-end UI Challenge that taps a Switch and asserts persisted+re-rendered state**. Five "reachable" screens have only `Class.forName` assertions (WEAK bluffs by §6.AB.3). Several screens have no UI test at all.

2. Highest-risk gaps (toggles / serialization / state-machines)

Ranked by the crash-class risk model (toggle/switch + serialization = highest, then state-machine flows, then static rendering):

R1 — provider_config Switches + Mirror/Clone serialization (HIGHEST — the just-found crash class)

feature/provider_config/sections/: SyncSection (Switch → ToggleSync), AnonymousSection (Switch → ToggleAnonymous), MirrorsSection (TextField → AddMirror/RemoveMirror/ProbeMirror), CloneSection + CredentialsSection + RemoveCloneSection (dialogs, bind/unbind).

- **Unit layer:** GREEN — ProviderConfigViewModelTest covers Finding-1 (ToggleAnonymous persistence across VM recreation), Finding-10 (ToggleSync DAO-flip race), AddMirror malformed-URL rejection + toast. Solid, falsifiability-stamped.
- **UI/Challenge layer: GAP.** Challenge04ProviderRow0OpensConfigTest only opens the screen and asserts "Sync this provider" is *displayed* — it never performClick()s the Switch, never asserts the persisted toggle survives, never exercises Mirror add / Clone create / Credential bind via the rendered UI. **The toggle/serialization crash class can only be caught end-to-end here, and no UI test does it.**

R2 — api-app Copy-key + OpenClient buttons (HIGH — serialization/clipboard + cross-app launch)

ApiControlScreen: CopyKeyClicked(key) (clipboard + key string), OpenClient (SiblingAppLauncher intent / Firebase fallback).

- C01–C05 GREEN-drive Start/Stop/Restart (via TAG_START/TAG_STOP test tags). **GAP:** no challenge taps Copy-key or Open-client and asserts the side effect.

R3 — search_input ProviderChipBar selection (HIGH — FilterChip toggle drives which providers get searched; prior bug: all-selected-by-default)

SearchInputScreen + components/ProviderChipBar.kt (FilterChips). **GAP** — **search_input has 0 direct challenge refs.** Search *result* flows are covered, but the chip-selection→submit boundary (a known prior defect surface) has only SearchInputViewModelTest at unit level.

R4 — onboarding state-machine (MEDIUM — well covered, keep as reference pattern)

GREEN. 7 challenges (C20–C29, C37, C41, C42) + 6 unit tests cover wizard flow, back-press, anonymous provider, password masking, finish-requires-proved-provider, select-all. This is the **reference quality bar** other features should match.

R5 — credentials / credentials_manager dialogs (MEDIUM — TextField + AlertDialog + secret handling)

CredentialEditDialog (TextField, save/cancel), CredentialsManagerScreen (TextField, dialogs). Unit tests exist (CredentialsViewModelTest, CredentialsManagerViewModelTest). UI: C47 (CredentialsLockedSearchSurvives) touches credentials indirectly. **WEAK/GAP** for the dialog edit→save→persist UI path and password-field masking in the manager.

R6 — rating dialog + menu sign-out (MEDIUM — dialog state-machine)

RatingDialog, MenuScreen sign-out AlertDialog. C30(rating reachable), C24(menu sign-out) exist — C24 GREEN (sign-out flow), C30 likely WEAK (reachable).

3. Coverage matrix (feature/screen × component × test status)

Legend: **G** GREEN (behavior asserted) · **W** WEAK (presence/classpath only) · — GAP.

Feature / Screen	Key interactive components	UI-test?	Unit-test?
onboarding (Welcome/ApiSelection/Configure/Providers/Summary)	TextFields, Switch (anon), provider toggles, nav	G (C20–29,37,41,42)	G (6 files)

Feature / Screen	Key interactive components	UI-test?	Unit-test?
provider_config (ProviderConfigScreen)	buttons, back-press Switch ×2 (Sync/Anon), TextField (mirror), AlertDialogs (clone/remove/bind)	W (C04 opens screen, asserts label only)	G (toggle pers AddMirror)
search (SearchScreen)	search box, nav	G (C02,03,09–12,38,40,44,46)	G
search_result (SearchResultScreen + Filter* + CrossTrackerFallbackModal)	FilterChips, dropdowns, fallback modal accept/dismiss	G (C07,08,16,39,43)	G (6 files)
search_input (SearchInputScreen + ProviderChipBar)	FilterChip provider selection, text field, submit	—	G (VM + nav roundtrip)
topic (TopicScreen)	download button, add-comment TextField, dialog	G (C05,06,37)	G (3 files)
login (LoginScreen/ ProviderLoginScreen/ OnboardingScreen)	TextFields, Switch, submit, captcha	G (C02,36)	G (4 files)
menu (MenuScreen)	nav rows, sign-out AlertDialog	G (C24 sign-out) W (C35 reachable)	G (3 files)

Feature / Screen	Key interactive components	UI-test?	Unit-test?
topic/connection (ConnectionItem/ ConnectionsList)	TextField, dialog		G (Connections)
category (CategoryScreen)	dialog, list nav	W (C34 nav reachable)	G
forum (ForumScreen)	list nav	partial G (C34?)	G
bookmarks (BookmarksScreen)	list, item actions	W (C33 reachable)	G
favorites (FavoritesScreen)	list, item actions	W (C32 reachable)	G
visited (VisitedScreen)	list, item nav	W (C31 Class.forName)	G
rating (RatingDialog)	dialog buttons	W (C30 reachable)	G
credentials (CredentialsScreen/ CredentialEditDialog)	TextField, AlertDialog save/cancel, secret field	partial (C47 indirect)	G
credentials_manager (CredentialsManagerScreen)	TextField, dialogs	—	G
account (AccountItem)	dialog	—	G (AccountVM)
main (MainScreen)	host/nav scaffold	— (covered transitively by launch challenges)	G (MainVM)
api-app (ApiControlScreen)	Start/Stop/ Restart buttons, Copy-key, Open-client	G (C01–05 drive Start/ Stop/Restart)	G (5 VM tests status labels)

4. Prioritized gap list

1. **provider_config toggle/serialization E2E UI (R1)** — tap Sync Switch → assert persisted + re-rendered; tap Anonymous Switch; AddMirror via TextField → assert row/toast in rendered UI; Clone

dialog create→confirm; Credential bind→unbind. *Crash class. No UI test exists.*

2. **search_input ProviderChipBar** (R3) — toggle provider chips → submit → assert only-selected providers searched (prior bug surface).
 3. **api-app Copy-key + Open-client** (R2) — drive both buttons, assert clipboard set / launch side effect.
 4. **5 WEAK "reachable" screens** (visited, favorites, bookmarks, category, connection) — upgrade from `Class.forName` to `render-and-interact`: open screen, assert empty-state OR list rows, tap an item, assert nav/side-effect.
 5. **credentials / credentials_manager dialogs** — edit-dialog `TextField`→save→persist; password masking in manager.
 6. **account, main, rating** — direct render + primary-action tests (account dialog; main scaffold renders nav; rating submit persists rating).
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5. Dispatch plan — 6 parallelizable waves

Each wave is independent (touches disjoint feature modules + disjoint `Challenge*Test.kt` files), so all 6 can run as parallel gap-filling agents. Every new test MUST: drive the real screen, primary-assert on user-visible state (§6.AB clause 1 rendering-correctness + clause 3 state-machine + clause 2 gating), and carry a §6.AB.3 non-crashing FALSIFIABILITY REHEARSAL block.

Wave 1 — provider_config toggle + serialization E2E (HIGHEST RISK)

- `Challenge48ProviderConfigSyncTogglePersistsTest` — open `ProviderConfig`, performClick the "Sync this provider" Switch, assert checked-state flips AND (recreate/return) the persisted value is reflected.
- `Challenge49ProviderConfigAnonymousTogglePersistsTest` — same for Anonymous switch.
- `Challenge50ProviderConfigAddMirrorUiTest` — type a valid URL in the mirror `TextField`, submit, assert the mirror row renders; type malformed URL, assert corrective toast + no row.
- `Challenge51ProviderConfigCloneDialogTest` — open clone dialog, enter name+URL, confirm, assert cloned provider appears.

- Falsifiability: revert each toggle handler to no-op; assert the new test fails on the re-render assertion.

Wave 2 — search_input chip selection (HIGH RISK, prior defect surface)

- Challenge52SearchInputProviderChipSelectionTest — render SearchInput, assert default-selected set matches onboarded providers (NOT all), toggle a chip off, submit, assert the de-selected provider is excluded from the search dispatch.
- Falsifiability: hardcode `selectedProviders = availableProviders`; assert test fails.

Wave 3 — upgrade 5 WEAK reachable screens to behavior tests

- Rewrite C31 (visited), C32 (favorites), C33 (bookmarks), C34 (category), C35 (connection) from `Class.forName` → `createAndroidComposeRule`, navigate to the screen, assert empty-state OR ≥ 1 list row renders, tap an item, assert the resulting nav/side-effect (user-visible).
- Falsifiability: break the nav entry; assert the render/empty-state assertion fails.

Wave 4 — api-app Copy-key + Open-client UI

- Challenge06ApiAppCopyKeyTest — boot+start, tap Copy on the access key, assert clipboard contains the key (or the confirmation side effect renders).
- Challenge07ApiAppOpenClientTest — tap "Open Lava client", assert the `LaunchClient` side effect intent is produced (installed vs Firebase fallback path).
- Falsifiability: make `CopyKeyClicked` a no-op; assert test fails.

Wave 5 — credentials + credentials_manager dialogs

- Challenge53CredentialEditDialogSavePersistsTest — open edit dialog, type into username/password TextFields, save, assert the credential persists and re-renders; assert password field is masked.
- Challenge54CredentialsManagerAddDeleteTest — add a credential via dialog, assert it lists; delete it, assert it's gone.
- Falsifiability: drop the save handler; assert test fails.

Wave 6 — account, main, rating direct render + primary action

- Challenge55AccountScreenRendersAndActsTest — render account, trigger its primary dialog/action, assert user-visible outcome.
 - Challenge56MainScaffoldRendersNavTest — assert mainScreen renders the bottom-nav entries (not transitive-only).
 - Challenge57RatingDialogSubmitPersistsTest — open rating dialog, select a rating, submit, assert persisted/side-effect (upgrade C30 from reachable→behavior).
 - Falsifiability per test as above.
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6. Notes for the dispatching stream

- **Onboarding (C20–29 etc.), search/search_result, topic, login, menu, api-app Start/Stop** are the GREEN reference patterns — copy their assertion style (`onNodeWithText(...).performClick()` → `waitUntil` → `assertIsDisplayed/persisted-row` check), NOT the `Class.forName` anti-pattern from C31–C35.
- All new client Challenge Tests share the §6.X-debt execution caveat: source-written + scanner-verified on darwin/arm64, EXECUTED on a Linux x86_64 + KVM gate-host (or per §6.AH container/VM path). State this honestly in each test's KDoc.
- The unit (ViewModel) layer is comparatively strong (38 client + 9 api-app test files). The gap is overwhelmingly at the **end-to-end rendered-UI layer**, which is exactly where the toggle/serialization crash class lives.